

Education

Nanjing University Sep 2023 – present
B.Eng. in Intelligent Software Engineering (expected Jun 2027)
School of Intelligent Software and Engineering
Average: 88.4/100

Research Interests

AI systems that learn and improve continually from data, experience, and human feedback. Interests: **data-centric AI, self-improving agents, and human–AI interaction.**

Research Experience

The Hong Kong University of Science and Technology (Guangzhou)
Visiting Research Student Aug – Oct 2026
Research with Prof. Weikai Yang

Nanjing University, iMATE Lab Mar 2025 – present
Research Assistant | Advisor: Prof. Ke Xu
School of Intelligence Science and Technology; collaboration with Dr. Wenchao Li.

Publications and Preprints

NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations
Wenchao Li, Yuewen Gao, Yu He, Cong Zhu, Ke Xu.
CHI 2026 | Published conference paper | DOI

SkillCommit: Evolving Agent Skills through Behaviorally Validated Scope Expansion
Yu He, Weikai Yang.
Submitted to AAAI 2027 | arXiv preprint, 2026 | Paper

MSSR: Memory-Aware Adaptive Replay for Continual LLM Fine-Tuning
Yiyang Lu, Yu He, Jianlong Chen, Hongyuan Zha.
arXiv preprint, 2026 | Paper

Manuscripts Under Review

SliceTune: Supporting Interactive Intent-Guided Slice Finding for Model Validation
Submitted to CHI 2027

SkillContour: Steering Agent Skill Consolidation with Behavioral Evidence
Submitted to CHI 2027

Selected Research Projects

SliceTune First author; research lead

- Designed and built a visual analytics system that maps validation goals to data attributes through a Feature Graph, letting practitioners steer failure-slice discovery.
- Designed and ran a 14-participant controlled study: users confirmed more failure slices across more attributes than the baseline in equal time.

SkillContour First author; research lead

- Designed and built a workbench that links source patches, skill instructions, and execution evidence to diagnose faulty consolidation and support regrouping, re-synthesis, and replay.
- Designed and ran a 14-participant study and 45-task held-out evaluation: users reported stronger diagnostic support than text-based review, and reviewed skills outperformed automatically consolidated and text-reviewed skills.

SkillCommit First author; research lead

- Designed and implemented skill consolidation using cross-instance replay and mechanism checks to assess compatibility, validating merged skills on source cases before commitment.
- Evaluated the method on RuleArena, OpenExempt, and KOR-Bench, showing improved agent performance without parameter updates and transfer of learned skills across model sizes and families.

NetworkCanvas Mar – Sep 2025

- Implemented visualization and feedback components for network exploration, contributing to the data integration and interactions behind adaptive recommendations and branching histories.
- Contributed to figures and study documentation; controlled studies found more noteworthy observations and higher analytical confidence than exploration without recommendations.

MSSR Aug 2025 – Jan 2026

- Modeled knowledge retention and implemented a replay scheduler that uses sample-level memory estimates to decide when to rehearse earlier examples.
- Integrated datasets and models for evaluation on three models and 11 sequential tasks; adaptive replay outperformed compared baselines, particularly on reasoning tasks.

NewsTract Jun 2025 – Jan 2026

- Designed data models and coordinated views linking events across news broadcasts through video, audio, and text.
- Contributed to a 16-participant study: users understood more events and identified more connections, with greater confidence than with the CCTV video archive.

Selected Honors and Awards

Nanjing University People’s Scholarship	Nov 2025
Meritorious Winner, Mathematical Contest in Modeling	May 2025
Consortium for Mathematics and Its Applications (COMAP).	
National Third Prize, China Collegiate Innovation and Invention Competition	Dec 2024

Technical Skills

Programming: Python, TypeScript, JavaScript, Java, C++, SQL.
Machine learning: PyTorch; continual fine-tuning and agent evaluation.
Interactive systems: React, Vue 3, D3.js, G6.js, ECharts; Git/GitHub.
Languages: Chinese (native); English.